

Problem Analysis Report

Undergraduate Transfer Management System

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Contents

1	Problem Context	5
1.1	Problem Overview	5
1.2	Problem Domains	5
1.2.1	Application Management Domain	5
1.2.2	Academic Evaluation Domain	5
1.2.3	Regulatory Compliance Domain	5
1.2.4	Approval Workflow Domain	5
1.2.5	Communication and Notification Domain	5
1.3	Jackson’s Context Diagram	5
2	AS-IS Process Definition	6
2.1	Process Overview	6
2.2	Stakeholder Interview Findings	6
2.3	Review of Policies, Procedures, and Guidelines	7
2.3.1	Relevant Documents	7
2.3.2	Key Policy Areas and Their Implications	7
2.4	AS-IS Process Definition (eEPC Model)	8
2.5	Sub-Process Definitions	10
2.5.1	Document Submission and Verification	10
2.5.2	Eligibility Assessment	10
2.5.3	Academic Evaluation and Ranking	10
2.5.4	Course Equivalence Evaluation	10
2.5.5	Sequential Approval Process	10
2.6	AS-IS Process Findings	10
3	TO-BE Process Definition	12
3.1	Purpose and Vision	12
3.2	System Boundaries	12
3.2.1	In-Scope Components	12
3.2.2	Out-of-Scope Components	12
3.2.3	System Constraints	13
3.3	Process Improvements	13
3.4	System Actors and Automation Scope	13
3.5	TO-BE Process Definition (eEPC Model)	14
3.6	Sub-Process Definitions	16
3.6.1	Application Verification (Automated)	16
3.6.2	Academic Evaluation (Semi-Automated)	16
3.6.3	Course Equivalence (System-Assisted)	16
3.6.4	Approval and Publication (Automated)	16
3.7	Differences Between AS-IS and TO-BE Models	16
3.8	Expected Outcomes	17
4	High-Level Functional Requirements	18
4.1	Application Management Requirements	18
4.2	Eligibility Verification Requirements	18
4.3	Academic Evaluation Requirements	18
4.4	Course Equivalence Requirements	19

4.5	Workflow and Approval Requirements	19
4.6	Dashboard and Reporting Requirements	19
4.7	Result Publication Requirements	20
4.8	Security and Audit Requirements	20
4.9	Integration Requirements	20
4.10	Performance and Usability Requirements	21
5	Conclusion	22

1 Problem Context

1.1 Problem Overview

The undergraduate transfer process at İzmir Institute of Technology (IZTECH) currently operates through a digital but still sequential system that involves multiple stakeholders and approval stages. Although the process is conducted digitally, it still results in significant delays, inconsistencies, and a lack of transparency for both applicants and administrative staff.

1.2 Problem Domains

The following domains are relevant to the undergraduate transfer problem:

1.2.1 Application Management Domain

This domain encompasses the submission, validation, and tracking of transfer applications. Currently, students submit their documents digitally through the system, but there is no integrated digital mechanism for tracking application status or receiving automated updates.

1.2.2 Academic Evaluation Domain

This domain involves GPA conversion, applicant ranking, and course equivalence (intibak) evaluation. Departmental Transfer Commissions (YGK) manually calculate weighted scores using spreadsheets, which is error-prone and time-consuming.

1.2.3 Regulatory Compliance Domain

This domain ensures adherence to YÖK (Council of Higher Education) regulations and IZTECH-specific transfer policies. Manual verification of eligibility criteria increases the risk of non-compliance and inconsistent application of rules.

1.2.4 Approval Workflow Domain

This domain manages the sequential approval process involving multiple organizational units: ÖİDB → Faculty Dean's Office → Departmental Transfer Commission → Faculty Board. Physical document circulation between these units creates bottlenecks.

1.2.5 Communication and Notification Domain

This domain handles information exchange between stakeholders. Currently relies on emails, phone calls, and physical memos, leading to delays and communication gaps.

1.3 Jackson's Context Diagram

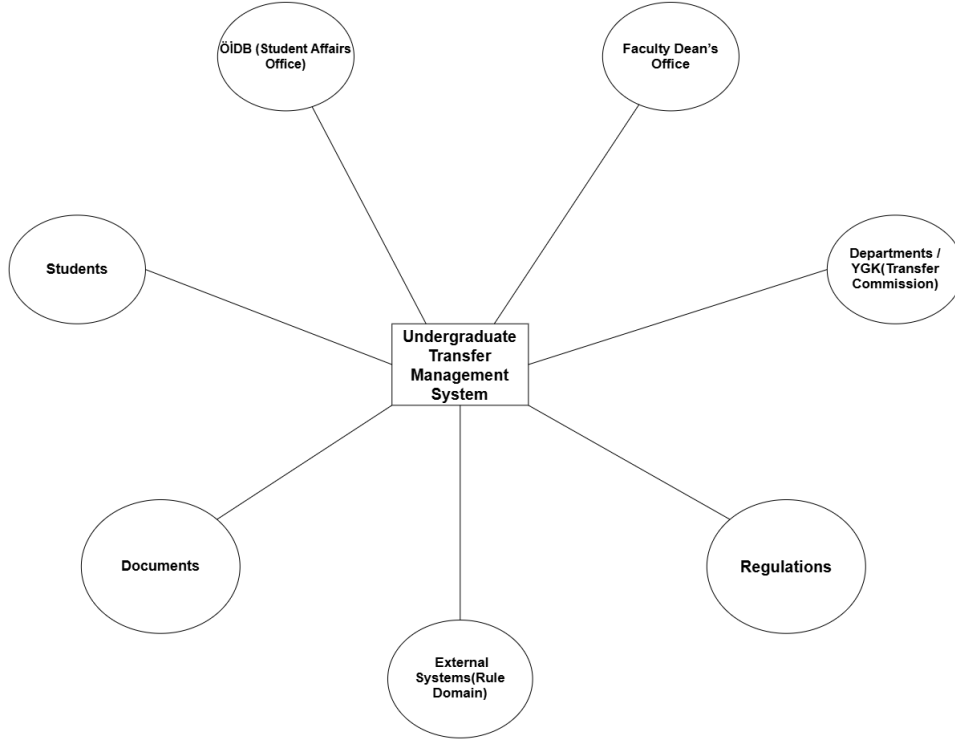


Figure 1: Jackson's Context Diagram for Undergraduate Transfer Management

2 AS-IS Process Definition

2.1 Process Overview

The current (AS-IS) undergraduate transfer process at IZTECH is a digital but still sequential workflow that relies on multiple approval steps and coordination between different stakeholders. Although the system operates digitally, the overall process still takes several weeks to complete.

2.2 Stakeholder Interview Findings

Interviews were conducted with the following stakeholders:

- **Students:** Expressed frustration about lack of visibility into application status and long processing times.
- **ÖİDB Staff:** Reported high workload in manual document verification and frequent inquiries from applicants.
- **YGK Members:** Highlighted time-consuming manual GPA calculations and difficulty in maintaining course equivalence tables.
- **Faculty Board Members:** Noted delays in receiving approval documents and limited time for thorough review.

2.3 Review of Policies, Procedures, and Guidelines

2.3.1 Relevant Documents

The following official documents regulate the undergraduate transfer process at IZTECH:

1. IZTECH Internal Undergraduate Transfer Regulation (07.07.2025) – Defines eligibility criteria, GPA thresholds, and course transfer (intibak) rules for students transferring within IZTECH.
2. IZTECH Inter-Institutional Undergraduate Transfer Regulation (21.12.2023) – Governs transfers from other universities to IZTECH and specifies the ranking and evaluation procedures.
3. IZTECH Regulation on Transfer with Central Placement Score (YKS-Based Transfer) – Covers transfers based on national university entrance scores.
4. IZTECH Application Documents for Internal and Inter-Institutional Transfers – Lists the required documents such as transcripts, exam results, and English proficiency certificates.
5. IZTECH Department-Specific Transfer Conditions (18.05.2023) – Defines additional departmental requirements, such as portfolio submission or minimum grades in core courses.

2.3.2 Key Policy Areas and Their Implications

Table 1: Key Policy Areas and AS-IS Process Implications

Area	Policy / Rule Summary	Implication for AS-IS Process
Documentation & Authentication	Applicants must submit official documents (transcript, YKS scores, English certificate, portfolio if required). All documents must be verified for authenticity.	ÖİDB manually checks document completeness and validity. Missing or incorrect documents delay processing.
Eligibility Criteria	Minimum GPA, successful completion of the first academic year, and compliance with YÖK's GPA conversion scale are mandatory.	YGK manually verifies GPA conversions and checks completion criteria.
Ranking and Evaluation	Applicant ranking is based on a weighted combination of GPA, entrance exam scores, and English proficiency.	YGK members manually calculate and rank applicants using spreadsheets or printed tables.

Area	Policy / Rule Summary	Implication for AS-IS Process
Course Equivalence (Intibak)	Course content, credits (ECTS), and prerequisites must be matched between institutions; final approval by Faculty Board is required.	YGK manually prepares and edits course equivalence tables; missing or unclear data often causes rework.
Approval and Announcement	Final results must be approved by Faculty Board and officially announced by ÖİDB.	Sequential approvals create delays in result publication and reduce transparency.
Department Specific Conditions	Certain departments impose additional conditions (e.g., design portfolio, minimum grade in key courses).	Special conditions introduce exceptions and additional manual checks.

2.4 AS-IS Process Definition (eEPC Model)

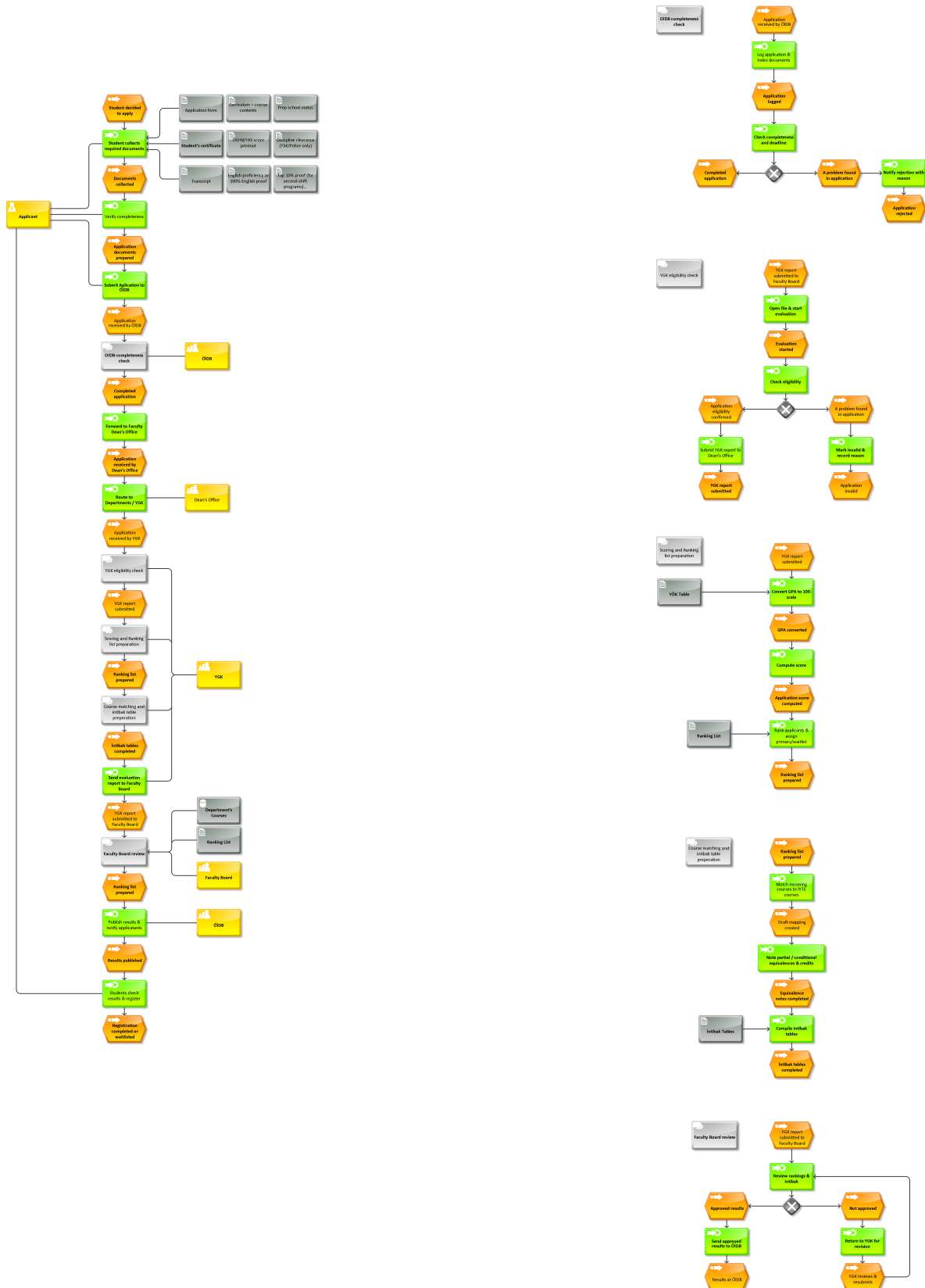


Figure 2: AS-IS eEPC Model for Undergraduate Transfer Process

2.5 Sub-Process Definitions

2.5.1 Document Submission and Verification

Actors: Student, ÖİDB Staff

Description: Students digitally submit their application documents to the Student Affairs Office (ÖİDB) through the online system. Staff reviews the submissions digitally to check completeness, authenticity, and format compliance. Incomplete applications are returned to students via the system for correction.

2.5.2 Eligibility Assessment

Actors: ÖİDB Staff, YGK Members

Description: Manual verification of minimum GPA requirements, semester completion, and compliance with YÖK regulations. Uses digitally updated forms and manual calculations.

2.5.3 Academic Evaluation and Ranking

Actors: YGK Members

Description: YGK members manually calculate weighted scores based on GPA and entrance exam results. Results are entered into spreadsheets for ranking.

2.5.4 Course Equivalence Evaluation

Actors: YGK Members

Description: Manual comparison of course syllabi, credits (ECTS), and content. Equivalence tables are created using word processors or spreadsheets.

2.5.5 Sequential Approval Process

Actors: Faculty Dean's Office, Faculty Board

Description: Physical documents are circulated for review and approval. Each level must sign and forward documents to the next level.

2.6 AS-IS Process Findings

Identified Shortcomings:

- **Digital Operations:** Document submission and verification are handled through a digital system; however, the process still experiences delays and inconsistencies due to sequential approvals and limited automation.
- **Manual GPA Evaluation:** Departments use spreadsheets for GPA calculations, prone to human error.
- **Lack of Transparency:** Students cannot monitor application status or see intermediate results.
- **Communication Bottlenecks:** Approval documents circulate physically between units, extending process time.

- **Compliance Risk:** Manual validation may overlook YÖK and institutional rules.
- **Limited Traceability:** No unified audit trail or version control for decisions.
- **Long Processing Time:** Complete process typically takes several weeks.
- **Resource Intensive:** Requires significant manual effort from all stakeholders.

Overall Assessment:

The AS-IS process strictly follows institutional regulations but heavily relies on manual verification and data entry, causing delays and possible human errors. Decision-making depends on coordination of multiple organizational units. The lack of digital integration between regulations and workflow steps leads to low efficiency and transparency.

3 TO-BE Process Definition

3.1 Purpose and Vision

The TO-BE model represents the automated and improved version of the Undergraduate Transfer Management System at IZTECH. It eliminates most of the manual evaluation process and introduces a digital workflow integrating all stakeholders into a unified web platform.

Primary Goals:

- Increase transparency, accuracy, and speed in application processing
- Maintain full compliance with institutional and YÖK regulations
- Provide real-time visibility for all stakeholders
- Reduce processing time
- Minimize human error through automation

3.2 System Boundaries

3.2.1 In-Scope Components

The following components are included within the system boundary:

- **Digital Application Management:** Online submission, document upload, and automatic validation of all required files (transcript, course contents, English proficiency, and exam results).
- **Regulation and Rule Validation:** Automatic enforcement of eligibility criteria defined by YÖK and IZTECH regulations, including GPA thresholds, semester limits, and department-specific conditions.
- **Automated Evaluation Workflow:** GPA conversion and ranking executed automatically by system algorithms and verified by departmental users.
- **Approval and Publication:** Faculty decisions digitally signed, and approved results instantly published to student portal.
- **Data Traceability and Audit Trail:** Every transaction logged for traceability and regulatory compliance.

3.2.2 Out-of-Scope Components

Course Equivalence and Adaptation: Course equivalence (intibak) evaluation and adaptation procedures are not automated within the system. All course matching and credit transfer decisions are manually performed by YGK in accordance with IZTECH Course Equivalence and Adaptation Regulation. The system only provides access to uploaded syllabi and transcripts to assist manual evaluation. Additionally, the evaluation and adaptation processes for international students are also outside the scope of this project.

3.2.3 System Constraints

- **Regulatory Constraints:** All workflows must fully comply with YÖK's Interinstitutional Transfer Regulation and IZTECH's internal transfer policies.
- **Technical Constraints:** The system's integrations depend on the availability of external services (ÖSYM, YÖKSİS). If these systems experience downtime or connection issues, certain validations may be delayed.
- **Organizational Constraints:** Final approval authority remains with departmental commissions (YGK) and faculty boards. The system supports and streamlines the process but **cannot replace or override their decisions.**

3.3 Process Improvements

Key Improvements from AS-IS:

- Centralized web portal integrating all actors
- Automated eligibility validation and GPA conversion
- Rule engine for YÖK and IZTECH regulation compliance
- Automated workflow triggers for each step
- Real-time notifications and dashboards
- Transparent, real-time result tracking
- Digital signature workflow
- Instant result publication

3.4 System Actors and Automation Scope

Table 2: System Actors and Their Automation Scope

Actor Role	Responsibilities	Automation Scope
Student (Applicant)	Uploads documents, tracks application status	Auto-validation, instant notifications, status dashboard
ÖİDB (Student Affairs)	Oversees process, verifies completeness, publishes results	Dashboard control, automated checks, one-click publication
Faculty Dean's Office	Monitors and approves digitally	Auto-forwarding, digital signature, notification system
Departments / YGK	Evaluates and ranks applicants, prepares equivalence tables	Auto GPA conversion, system-generated ranking, document access
Faculty Board	Reviews and provides final approval	E-signature, digital review interface, approval workflow

Actor Role	Responsibilities	Automation Scope
YDYO	Validates English proficiency documentation	Automated verification of language exam results when integrated
External Systems	Provide validation data	API integration (ÖSYM, YÖKSİS) for automated verification

3.5 TO-BE Process Definition (eEPC Model)

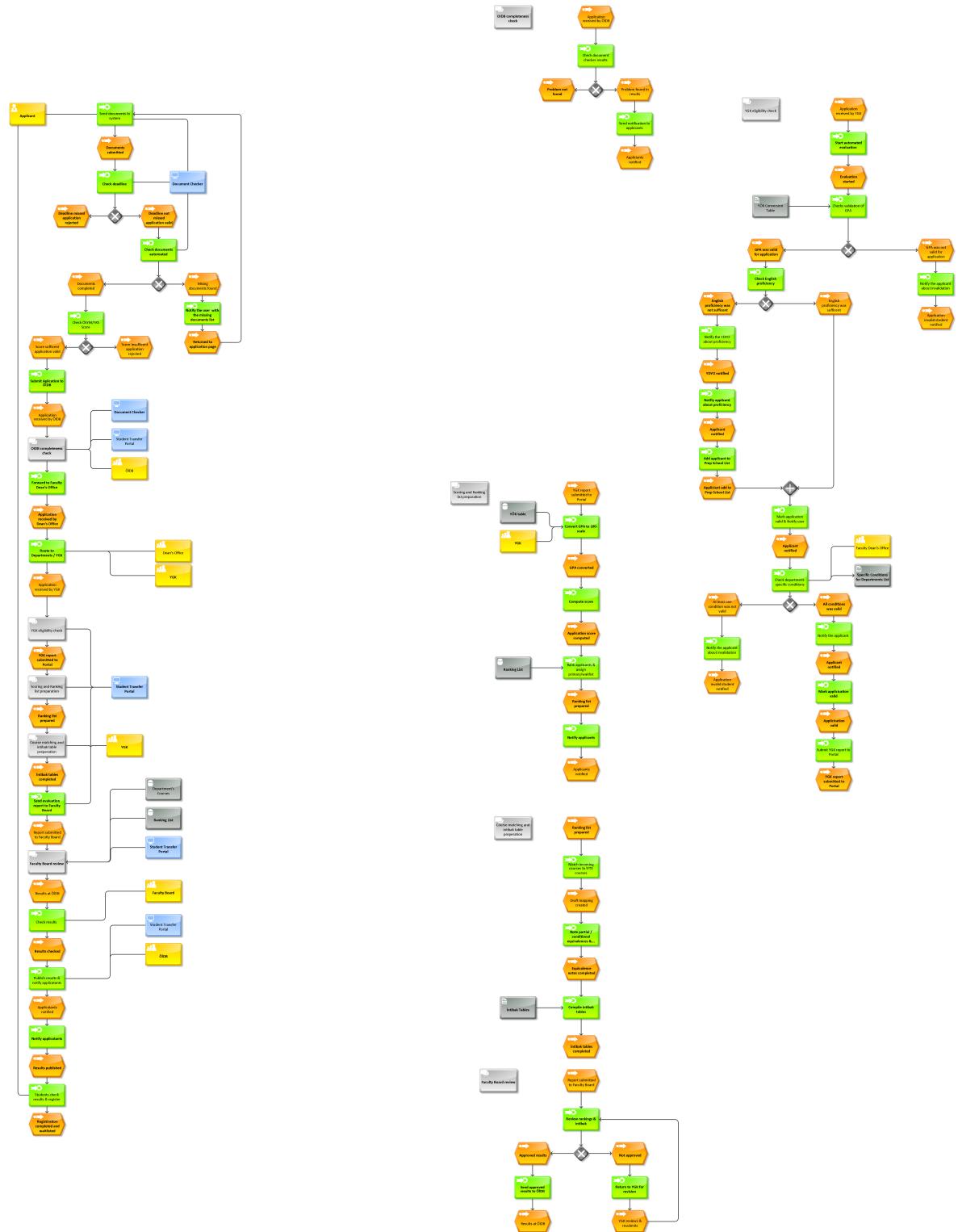


Figure 3: TO-BE eEPC Model for Undergraduate Transfer Process

3.6 Sub-Process Definitions

3.6.1 Application Verification (Automated)

Actors: Student, System, ÖİDB

Description: System automatically checks document formats and authenticity through API validation with ÖSYM and YÖKSİS. Eliminates manual inspection. Real-time feedback provided to student.

3.6.2 Academic Evaluation (Semi-Automated)

Actors: YGK Members, System

Description: System automates GPA conversion according to YÖK standards and generates applicant ranking. YGK reviews and verifies system-generated results. Replaces manual spreadsheets with centralized scoring interface.

3.6.3 Course Equivalence (System-Assisted)

Actors: YGK, System

Description: System provides organized access to uploaded syllabi and transcripts. YGK performs manual evaluation using system interface. Previous equivalence decisions stored for reference.

3.6.4 Approval and Publication (Automated)

Actors: Faculty Board, ÖİDB, System

Description: Digital signature workflow with automated routing. Instant result publication to student portal upon final approval. Email notifications sent automatically to all stakeholders.

3.7 Differences Between AS-IS and TO-BE Models

Table 3: Comparison of AS-IS and TO-BE Models

Area	AS-IS Model	TO-BE Model
Document Handling	Digital submission with manual verification	Digital submission with automated validation via APIs
Eligibility Check	Manual verification by ÖİDB and YGK	Automated rule-based validation by system
GPA Calculation	Manual spreadsheet calculations	Automated conversion using YÖK algorithms
Ranking	Manual sorting in spreadsheets	System-generated ranking with verification option
Course Equivalence	Manual table creation in word processors	System-assisted evaluation with document access
Approval Flow	Sequential physical document circulation	Parallel digital approvals with e-signatures
Communication	Emails, phone calls, physical memos	Instant dashboard notifications and alerts

Area	AS-IS Model	TO-BE Model
Transparency	No visibility until final results	Real-time status tracking for all stakeholders
Processing Time	Several weeks (15-20 working days)	Several hours to 2-3 days
Error Rate	High (manual data entry and calculations)	Minimal (automated validation and calculation)
Audit Trail	Limited or absent	Complete transaction log with timestamps
Scalability	Limited by manual capacity	Highly scalable with automated processes

3.8 Expected Outcomes

- **Time Efficiency:** Processing time reduced
- **Accuracy:** Human errors are minimized through automation and validation
- **Transparency:** Students and administrators have real-time visibility of application status.
- **Compliance:** The system enforces regulatory rules automatically.
- **Traceability:** A complete audit trail is maintained for all transactions and decisions.
- **User Satisfaction:** Stakeholders experience improved usability and response times.
- **Resource Optimization:** Administrative staff workload is reduced.
- **Consistency:** Evaluation and decision-making processes are standardized.

4 High-Level Functional Requirements

Based on the TO-BE process model, the following high-level functional requirements have been identified for the Undergraduate Transfer Management System:

4.1 Application Management Requirements

1. The system shall allow students to create and submit transfer applications online.
2. The system shall allow students to upload required documents (transcript, course syllabi, English proficiency certificate, exam results) in PDF format.
3. The system shall automatically validate uploaded documents for format compliance and completeness.
4. The system shall integrate with ÖSYM and YÖKSİS APIs to verify document authenticity.
5. The system shall provide real-time application status tracking to students.
6. The system shall send automatic email notifications to students upon status changes.
7. The system shall allow students to edit and resubmit applications before the deadline.

4.2 Eligibility Verification Requirements

1. The system shall automatically verify minimum GPA requirements based on YÖK and IZTECH regulations.
2. The system shall check semester completion requirements automatically.
3. The system shall validate department-specific eligibility criteria.
4. The system shall prevent submission of applications that do not meet basic eligibility criteria.
5. The system shall generate eligibility reports for ÖİDB review.
6. The system shall maintain a rules engine for configurable eligibility criteria.

4.3 Academic Evaluation Requirements

1. The system shall automatically convert GPAs according to YÖK's standardized conversion scale.
2. The system shall calculate weighted scores based on GPA, entrance exam results, and English proficiency.
3. The system shall automatically rank applicants within each department based on calculated scores.
4. The system shall allow YGK members to review and verify automated calculations.

5. The system shall allow YGK members to manually adjust rankings if necessary with justification.
6. The system shall generate comparison reports showing all applicants' scores side-by-side.

4.4 Course Equivalence Requirements

1. The system shall provide YGK members access to all uploaded course syllabi and transcripts.
2. The system shall allow YGK members to create and edit course equivalence tables digitally.
3. The system shall store historical course equivalence decisions for reference.
4. The system shall allow YGK to mark courses as equivalent, partial, or non-equivalent.
5. The system shall generate equivalence reports for Faculty Board review.

4.5 Workflow and Approval Requirements

1. The system shall automatically route applications to appropriate departments based on student's target program.
2. The system shall support digital signature functionality for all approval stages.
3. The system shall allow parallel review and approval by multiple stakeholders when applicable.
4. The system shall enforce sequential approval workflow: ÖİDB → YGK → Faculty Board.
5. The system shall send automatic notifications to responsible parties when action is required.
6. The system shall allow approvers to add comments and feedback at each stage.
7. The system shall support approval delegation when primary approvers are unavailable.

4.6 Dashboard and Reporting Requirements

1. The system shall provide role-based dashboards for each user type (student, ÖİDB, YGK, Faculty Board).
2. The system shall display pending tasks and notifications prominently on dashboards.
3. The system shall generate statistical reports on application volumes, processing times, and outcomes.
4. The system shall provide real-time monitoring of overall process progress for administrators.

5. The system shall allow export of reports in PDF and Excel formats.
6. The system shall visualize application statistics using charts and graphs.

4.7 Result Publication Requirements

1. The system shall automatically publish approved transfer results to the student portal.
2. The system shall send email notifications to all applicants when results are published.
3. The system shall display detailed feedback for rejected applications.
4. The system shall allow ÖİDB to schedule result publication for specific date and time.
5. The system shall maintain published results in the system archive.

4.8 Security and Audit Requirements

1. The system shall implement role-based access control for all users.
2. The system shall log all user actions with timestamps for audit trail.
3. The system shall encrypt sensitive student data both in transit and at rest.
4. The system shall require multi-factor authentication for administrative users.
5. The system shall maintain version history for all documents and decisions.
6. The system shall allow authorized users to view complete audit logs.
7. The system shall comply with KVKK (Turkish Personal Data Protection Law) requirements.

4.9 Integration Requirements

1. The system shall integrate with ÖSYM API for entrance exam score verification.
2. The system shall integrate with YÖKSİS API for transcript verification.
3. The system shall support standard document formats (PDF, DOCX) for uploads.
4. The system shall provide API endpoints for potential future integrations.
5. The system shall handle external API failures gracefully with appropriate error messages.

4.10 Performance and Usability Requirements

1. The system shall support concurrent access by at least 500 users.
2. The system shall respond to user actions within 3 seconds under normal load.
3. The system shall be accessible via modern web browsers (Chrome, Firefox, Safari, Edge).
4. The system shall be responsive and usable on mobile devices (tablets and smart-phones).
5. The system shall provide help documentation and user guides.
6. The system shall support both Turkish and English languages.
7. The system shall maintain 99% uptime during application periods.

5 Conclusion

This problem analysis report has comprehensively examined the undergraduate transfer process at İzmir Institute of Technology, analyzing both the current (AS-IS) manual process and the proposed (TO-BE) automated system.

Key Findings:

- The AS-IS process, while compliant with regulations, suffers from inefficiency, lack of transparency, and high error potential due to its manual, paper-based nature.
- The TO-BE process addresses these shortcomings through comprehensive automation, digital workflows, and integration of all stakeholders into a unified platform.
- The proposed system is expected to reduce processing time, minimize human error, and provide real-time visibility for all parties.
- High-level functional requirements have been identified covering all aspects of the system, from application submission to result publication.

The transformation from the AS-IS to TO-BE model represents a significant modernization of the transfer process, aligning İZTECH with contemporary digital transformation practices in higher education while maintaining strict compliance with YÖK and institutional regulations.

The next phases of the project will involve detailed system design, development, testing, and deployment of the Undergraduate Transfer Management System based on the requirements and process models defined in this report.

Note on Review Comments:

Rejected Items:

1. **3.3 Process Improvements:** We did not get what the review team meant when they remarked 'it is insufficient.' We do not think there is no connection with AS-IS.
2. **3.2.2 Course Equivalence and Adaptation:** The review team stated that Course Equivalence and Adaptation part is included both within and outside the scope, but it is only included in the out-of-scope part. It is also stated by the review team that even though this section is described as out of scope, it appears back in 4.4. But 4.4 only specifies what degree of control will be provided to the YGK members and has nothing to do with evaluating course equivalence which is stated as out-of-scope.
3. **2.4 (AS-IS eEPC Model):** The review team noted that "&" means "and" in the diagram. Additionally, they mentioned there is logging and indexing in the system. The level of detail provided in the AS-IS eEPC model was deemed sufficient by the reviewers.
4. **3.5 (TO-BE eEPC Model):** The review team confirmed that the intelligibility of the TO-BE eEPC model is sufficient. They also noted that "&" means "and" in the diagram, and the level of detail provided is adequate.